

Positive Blood Culture Interpretation

The best way to prevent blood culture contamination is to only collect blood cultures when indicated (e.g., septic shock, suspected endovascular infection). Patients who do not display systemic signs of infection are unlikely to have bacteremia in most circumstances.

Blood cultures should be collected from clean, prepped skin at >1 site. Blood cultures should NOT be collected off an IV except when the diagnosis of catheter associated bloodstream infection is suspected.

Next step based on Gram stain interpretation: Call micro lab at x55148 to obtain BioFire BCID2 species result. If listed as Staph spp. or not identified on BioFire, proceed to pathway #2 below:

1. If *Staphylococcus aureus*, *Staphylococcus lugdunensis*, *Listeria* spp. or Yeast -> Call back to ER, repeat culture prior to starting antibiotics, consult ID
2. If common skin contaminant: *
 - a. If central line, prosthetic valve, endovascular cardiac device, or scenario/other risk factors for infective endocarditis -> Call back to ER, repeat cultures
 - b. Otherwise, use clinical judgement with low threshold to call back to ER to recollect blood cultures
3. If any other Gram-positive cocci or Gram-negative rod -> Call back to ER for further workup and evaluation

Refer to [VA Ann Arbor BCID Guidance](#) for recommended therapies.

A positive blood culture is more likely to be a contaminant if:

1. Organism is skin flora* (see list below)
2. Only isolated from one of multiple sets
3. Time-to-positivity > 20 hours
4. Low suspicion for bacteremia (e.g., afebrile, no systemic signs of infection)

***Common blood culture contaminants:**

1. Coagulase-negative staphylococci (e.g., *capitis*, *hominis*, *epidermidis*, *simulans*, *haemolyticus*)
2. *Bacillus* spp. (except *B. anthracis*)
3. *Corynebacterium* spp.
4. *Propionibacterium* (*Cutibacterium*) spp.
5. *Micrococcus* spp.

*All organisms on this list have potential to cause true disease.

**Please reach out to ID (via consult, or page ID fellow at #3010)
to discuss individual cases at any time**